

Laser-constrained control of molecular formation

A history-dependent two-channel optimisation through an optical Feshbach resonance

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THE QUESTION

How do physical laser constraint and power cost limits affect maximum molecular yield, and how robust is our optimised yield calculation?

Physical control problem

An optical Feshbach resonance is a laser-induced resonance between two colliding atoms in an open scattering channel and an optically excited molecular bound state in a closed channel. In the photon-dressed energy picture, the molecule and an absorbed laser photon are treated as one combined state. Changing the laser frequency then moves its effective bound-state energy E_c relative to the open-channel scattering energy E . Strong resonance requires $E_c - E \approx 0$; for ultralow threshold collisions $E \approx 0$, so resonance requires $E_c \approx 0$.

Once the channels are near resonance, detuning ν controls the remaining energy mismatch, while laser intensity sets the channel coupling $W(I)$ and stimulated width $\Gamma \propto |W(I)|^2$. Because the excited molecule decays at rate γ , the resonance produces a *complex* scattering length: the dispersive interaction and inelastic molecular flux are linked responses, not independent controls.

The conversion rate is also nonlocal in time. A short-range pair amplitude $\eta(t)$ retains the complete earlier scattering history, so optimising a constant laser setting is not equivalent to optimising a pulse.

Coupling-loss constraint

Near resonance, the closed-channel fraction scales approximately as

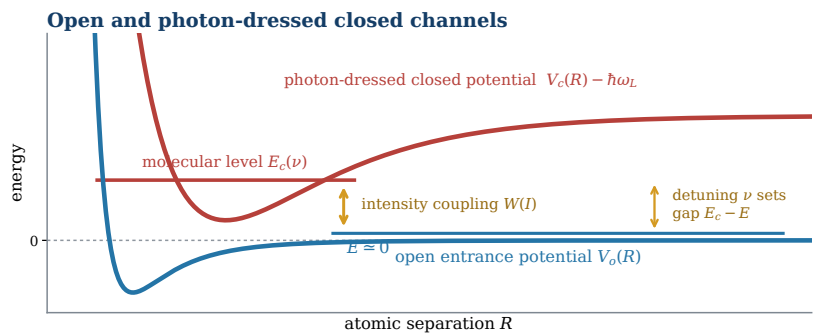
$$|b|^2 \propto \frac{|W(I)|^2}{(E - E_c)^2 + (\hbar\gamma/2)^2}.$$

Detuning suppresses both admixture and decay; intensity strengthens both. The useful operating path must therefore be found dynamically.

Physical reference: ^{88}Sr

The 689 nm $^1S_0 - ^3P_1$ line gives $a_{\text{bg}} = -1.4(6)a_0$. An ideal excited molecule has $\gamma/2\pi = 15$ kHz, whereas the direct OFR experiment measured an effective 40(5) kHz width. Its isolated-line model works only close to one molecular level [8, 9].

A driven two-channel resonance



We denote the open and photon-dressed closed potentials by $V_o(R)$ and $V_c(R) - \hbar\omega_L$, respectively. The former supports a near-zero-energy scattering state, while the latter supports a molecular bound state E_c . Detuning controls the mismatch $E_c - E$ and intensity sets the coupling $W(I)$; efficient transfer requires a small mismatch and non-zero coupling. Concept adapted to the optical case from the two-channel construction of Chin *et al.* [7].

Assumed dimensional scattering length. The one-pole model used by the calculation is

$$a_c(I, \nu) = a_{\text{bg}} \left[1 + \frac{s\Gamma(I)}{-\nu - s\Gamma(I) + i\gamma/2} \right], \quad s = \text{sgn}(a_{\text{bg}}), \quad \Gamma(I) = \kappa I \geq 0.$$

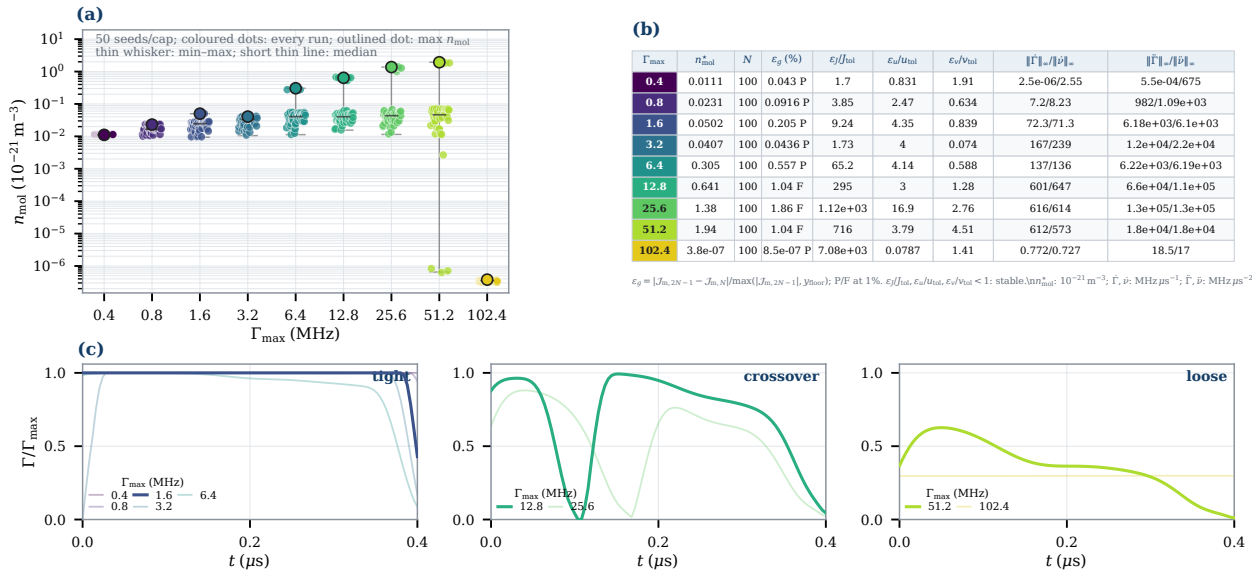
Here Γ , ν , and γ use the same frequency units. For ^{88}Sr , $a_{\text{bg}} = -1.4a_0$ gives $s = -1$, which preserves the loss sign $\text{Im } a_c < 0$.

Assumptions: low-energy s -wave scattering with $E \rightarrow 0$; one open channel and one isolated excited molecular state; renormalised zero-range coupling; constant Markovian molecular decay γ ; weak-field scaling $\Gamma = \kappa I$; and the tied light shift $\nu_{\text{re}} = \nu + s\Gamma$. The last identification is a model parameterisation, not a universal OFR identity [11].

Current numerical evidence

Completed cap and regularisation sweeps, followed by targeted refinement, expose both reproducible low-cap structure and an unresolved high-cap instability.

PRELIMINARY
VALIDATION IN PROGRESS



Data snapshot 2026-08-10 02:27 UTC. Density remains a counting-convention-dependent proxy.

Figure 2. Current numerical evidence (snapshot 10 August 2026 UTC). (a) n_{mol} for all 450 seeds at all 9 Γ_{max} values; whiskers span min-max, short lines give medians, and outlined points give the maximum density. (b) Per-cap audit of that run: N , nested-grid error ϵ_g , terminal stability ratios $\epsilon_j/J_{\text{tol}}$, $\epsilon_u/u_{\text{tol}}$, $\epsilon_v/v_{\text{tol}}$, and dimensional first/second control derivatives. Here $\epsilon_g = |\mathcal{J}_{m,2N-1} - \mathcal{J}_{m,N}| / \max(|\mathcal{J}_{m,2N-1}|, y_{\text{floor}})$; P/F denotes the 1% test. (c) Regularised $\Gamma/\Gamma_{\text{max}}$ histories against physical t , grouped as tight, crossover, and loose. Caps 40, 320, and 1280 are fully opaque; the other cap histories remain visible but faded.

450

fixed-method seeds
displayed

9×50

caps × seeds per cap

0.0502 / 0.582

n_{mol} proxy in 10^{-21} m^{-3} at
1.6/12.8 MHz

0/10

strict runs meeting stop
criteria

What is reproducible

Fixed campaign. One 111,000-step Adam schedule produced 450 runs at $N = 100$. Panel (a) keeps every seed at all 9 caps, including collapsed solutions.

Tight cap. Independent refinements repeatedly give $n_{\text{mol}} \approx 0.0502 \times 10^{-21} \text{ m}^{-3}$ at $\Gamma_{\text{max}} = 1.6 \text{ MHz}$; the control remains intensity-limited.

Intermediate cap. A 250-seed screen, all-seed polish, and top-20 L-BFGS refinement give $0.582 \times 10^{-21} \text{ m}^{-3}$ at $\Gamma_{\text{max}} = 12.8 \text{ MHz}$. The endpoint is time-limited.

High-cap instability

Heavy tail. At $\Gamma_{\text{max}} = 51.2 \text{ MHz}$, the median is only $0.0464 \times 10^{-21} \text{ m}^{-3}$ while the best seed reaches $1.94 \times 10^{-21} \text{ m}^{-3}$. At 102.4 MHz every seed collapses; neither behaviour defines saturation.

Refinement still climbs. The latest trajectory reaches $n_{\text{mol}} = 5.34 \times 10^{-21} \text{ m}^{-3}$ after 10,605,750 steps. Transfer from $N = 100$ to 800 changes it by -5.43% , but matched reoptimisation remains outstanding. Narrow grid-scale excursions therefore cannot yet be interpreted as an available yield enhancement.

Interpretation and validation

Panel (c) compares $\Gamma/\Gamma_{\text{max}}$ on the physical $0.4 \mu\text{s}$ horizon. The highly refined cap-40, cap-320, and cap-1280 solutions are emphasised; the remaining fixed-sweep controls provide faded context.

Panel (b) reports the stored $N \rightarrow 2N - 1$ forward check, all three terminal stability ratios, and both derivative orders for Γ and ν . Matched reoptimisation is still required before claiming a crossover cap or plateau.

Absolute molecule number also awaits reconciliation of the initialisation, kernel coefficient, and counting prefactor.

PART II / TECHNICAL REPORT AND DERIVATION

Optimal Laser-Constrained Control of Molecular Formation from a Quantum Gas via Optical Feshbach Resonances

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Abstract

Optical Feshbach resonances couple an ultracold entrance channel to a decaying molecular state, so laser intensity and detuning jointly control both dispersive interactions and inelastic conversion. We formulate this as a bounded optimal-control problem: the laser determines a complex scattering length, a half-order Volterra equation propagates the history-dependent pair amplitude, and the loss-contact relation defines a dimensionless accumulated molecular objective. Automatic differentiation through this nonlocal dynamics permits multi-start optimisation of the complete intensity and detuning histories. A completed $N = 100$ campaign contains 450 runs over 9 intensity caps, with 50 starts per cap; the showcase displays every seed at every cap. Later searches give a stable saved value near $\mathcal{J}_m = 81.68$ at cap 40 and a time-limited selected endpoint $\mathcal{J}_m = 947.1$ at cap 320. At cap 1280, however, the objective continues to grow through increasingly grid-scale controls. The current calculation therefore establishes a power- and methodology-sensitive numerical landscape, not a converged high-power optimum or absolute molecule yield. Kernel normalisation, initialisation, time-grid convergence, the reduced optical map, and the experimental counting prefactor remain explicit validation requirements.

Keywords: optical Feshbach resonance, ultracold quantum gas, Tan contact, molecular formation, optimal control, Volterra equation, JAX

Data and validation status (10 August 2026 UTC). Numerical statements in the Results section are regenerated from immutable configuration-document and queue identifiers; the accompanying JSON file records every source. Figure 2 uses the stated ^{88}Sr conversion for n_{mol} , Γ_{max} , t , and the control derivatives; only the explicitly requested ratio $\Gamma/\Gamma_{\text{max}}$ is dimensionless. Items marked **[TO COMPLETE]** identify unresolved derivation, matched-grid reoptimisation, or experimental-calibration work and are not hidden by the updated presentation.

1. INTRODUCTION

Feshbach resonances make the interaction strength of an ultracold gas a controllable experimental parameter. At low collision energy, this control is summarised by the s -wave scattering length, and resonant coupling can change both its magnitude and sign [7]. Optical coupling adds speed and spatial selectivity, including for species without useful magnetic resonances, at the cost of spontaneous-emission and photoassociation loss [1, 5].

The central difficulty is dynamical. Molecular production at one instant depends not only on the laser settings at that instant, but also on the short-range pair correlations accumulated over the entire preceding protocol. Tan's contact organises these correlations [4, 6], while the universal short-time theory of Qi, Shi, and Zhai reduces their evolution to a history-dependent equation for a pair amplitude $\eta(t)$ [10]. This combination turns molecular formation into an optimal-control problem with memory.

Central research question

For a fixed observation time T and a maximum available laser coupling Γ_{max} , which experimentally realisable intensity and detuning trajectories maximise the accumulated molecular yield, and when does increasing Γ_{max} cease to improve that yield?

1.1. Motivation and scope

Abstract control of the real and imaginary parts of a complex scattering length can generate protocols that no laser can realise. This work instead uses the physical laser variables $\Gamma(t)$ and $\nu(t)$ and enforces $0 \leq \Gamma(t) \leq \Gamma_{\text{max}}$. A smoothness penalty and a posteriori slew-rate tests distinguish numerically optimal pulses from controls that can plausibly be produced in the laboratory.

The intended output is a cap-response curve: optimised yield as a function of $\Gamma_{\text{max}}/\gamma$, spanning the tight-cap and saturation regimes.

The physical representative is fixed to bosonic ^{88}Sr , using the 1S_0 - 3P_1 intercombination line near 689 nm and an excited Sr_2 photoassociation level. This choice supplies measured linewidth, optical-length, cloud, and laser scales for translating the dimensionless calculation [8, 9].

1.2. Objectives and contributions

This report is organised around the following deliverables:

1. verify the optical two-channel mapping and its limiting cases;
2. validate a discrete solver for the half-order Volterra dynamics;
3. optimise bounded physical controls with a reproducible multi-start procedure;
4. calibrate the learning-rate schedule and control smoothness, then test convergence, grid dependence, and seed sensitivity;
5. identify cap-binding and cap-insensitive regimes in the molecular yield; and
6. translate one optimised dimensionless protocol into experimentally meaningful intensity, detuning, and duration.

The present work completes the first cap-response campaign and two generations of regularisation search. The tight-cap result is reproducible across several optimiser stages, while the intermediate-cap value improves under targeted refinement. The principal new finding is negative but consequential: at high cap, longer optimisation increases $\mathcal{J}_m$ by creating structure on the $N = 100$ time scale, so the apparent upper envelope is not grid validated. The study therefore narrows the trustworthy operating region and specifies the resolution tests needed before an optical-power optimum can be claimed.

2. PHYSICAL BACKGROUND

2.1. Two-channel picture

An optical Feshbach resonance is a laser-induced resonance between an open atomic scattering channel and an optically excited molecular bound state in a closed channel. The two-body state is consequently decomposed as

$$|\Psi\rangle = |\psi_o\rangle|o\rangle + |\psi_c\rangle|c\rangle. \quad (1)$$

When a laser couples these channels with strength W , the states mix. Near an isolated closed-channel energy level E_c , write $|\psi_c\rangle \simeq b|\phi_c\rangle$. Projecting the stationary Schrödinger equation onto this level gives the closed-channel amplitude

$$b = \frac{\langle\phi_c|W^\dagger|\psi_o\rangle}{E - E_c + i\hbar\gamma/2}, \quad (2)$$

where γ is the decay rate of the excited molecule. The closed-channel admixture, which scales as $P_c \propto |b|^2$, is therefore enhanced near resonance but suppressed by large energy detuning or weak coupling. This mechanism provides the microscopic origin of the resonant, lossy control used below [7].

2.2. Low-energy scattering

The open-channel dynamics at low energy are governed by the s -wave scattering length a_s . Outside the interaction range, the zero-energy scattering wavefunction takes the asymptotic form

$$\psi_o(r) \propto 1 - \frac{a_s}{r}. \quad (3)$$

Purely elastic scattering yields a real scattering length. The molecular decay γ introduced in the two-channel picture instead removes flux irreversibly, requiring a complex scattering length. With the time-dependence and $+i\gamma/2$ denominator convention used by the implementation,

$$a_s = \alpha - i\beta, \quad \beta \geq 0. \quad (4)$$

The real part α controls the elastic phase shift, while β captures irreversible flux into molecular or radiative channels [1]. Consequently $\text{Im}(1/a_s) \geq 0$ wherever $a_s \neq 0$. We employ a zero-range model that enforces this boundary condition while neglecting finite-range microscopic structure inside the interaction region.

[TO COMPLETE] State the regime in which the zero-range and short-time approximations are valid for the chosen gas. Define all density, range, and many-body timescale inequalities that will later be checked.

2.3. Short-time pair amplitude

Consider two identical atoms of mass m in their relative coordinate $\mathbf{r}$. At times short compared with the many-body timescale, their s -wave dynamics are governed by the zero-range Schrödinger equation with the Fermi–Huang pseudopotential:

$$i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2}{m} \nabla^2 + \frac{4\pi\hbar^2}{m} a_s(t) \delta^{(3)}(\mathbf{r}) \frac{\partial}{\partial r} r \right] \psi. \quad (5)$$

We assume an initially uniform, noninteracting state,

$$\psi(\mathbf{r}, 0) = \frac{1}{\sqrt{\mathcal{V}}}, \quad a_s(0) = 0, \quad (6)$$

where $\mathcal{V}$ is the normalisation volume. Thus the initial position-space probability density is uniform, $|\psi(\mathbf{r}, 0)|^2 = 1/\mathcal{V}$.

To solve the short-time dynamics, absorb the time-dependent contact boundary condition into the length-valued short-range amplitude

$$\eta(t) = -4\pi a_s(t) \sqrt{\mathcal{V}} \lim_{r \rightarrow 0} \frac{\partial}{\partial r} [r\psi(r, t)]. \quad (7)$$

This definition turns the interaction into a time-dependent point source at the origin. Applying Duhamel’s formula with the free-particle Green function separates the solution into a $1/r$ spatial singularity and a regular, history-dependent remainder. This finite history

is governed by the half-order operator [10]

$$\mathcal{L}_{1/2}\eta(t) = c \lim_{\epsilon \rightarrow 0^+} \left[\int_0^{t-\epsilon} \frac{\eta(\tau)}{(t-\tau)^{3/2}} d\tau - \frac{2\eta(t)}{\sqrt{\epsilon}} \right], \quad (8)$$

where

$$c = \sqrt{\frac{m}{\hbar}} \frac{1}{8\pi^{3/2}\sqrt{i}}. \quad (9)$$

Evaluating the spatial derivative of the Green-function solution near the origin recovers the boundary condition and closes the dynamics as a single Volterra integral equation for $\eta(t)$:

$$\left[\mathcal{L}_{1/2} + \frac{1}{4\pi a_s(t)} \right] \eta(t) = -1. \quad (10)$$

Once $\eta(t)$ is determined, it fixes the short-distance form of the wavefunction through the Bethe–Peierls boundary condition

$$\psi(\mathbf{r}, t) = \frac{\eta(t)}{4\pi\sqrt{\mathcal{V}}} \left[\frac{1}{r} - \frac{1}{a_s(t)} \right] + \mathcal{O}(r). \quad (11)$$

Because the system starts noninteracting, $\eta(0) = 0$, and its subsequent evolution encodes the full history of the controlled scattering length.

2.4. Tan’s contact and universal short-time dynamics

Tan’s contact $\mathcal{C}$ quantifies short-range pair correlations and appears, for example, in the large-momentum tail

$$n_g(k) \xrightarrow{k \rightarrow \infty} \frac{\mathcal{C}}{k^4}. \quad (12)$$

It connects microscopic pair structure to thermodynamic and dynamical observables [4, 6]. For a gas initially in a noninteracting state, the universal short-time reduction used here writes

$$\mathcal{C}(t) = g_2(0) |\eta(t)|^2, \quad (13)$$

where $g_2(0)$ is the initial local pair density and $\eta(t)$ is determined by two-body short-distance physics [10].

A real square-root ramp, $a_s(t) \propto \sqrt{\hbar t/m}$, gives the scale-invariant onset studied in Ref. [10]. It provides both a conceptual benchmark and a numerical limiting case for the present extension to complex a_s .

2.5. Laser control of the scattering length

Up to this point, $a_s(t)$ is an arbitrary prescribed interaction history. For the optical-control problem, Eq. (10) is analytically continued to a complex scattering length and that history is restricted to values accessible through laser intensity and detuning. A “photon-dressed” closed channel is the coupled molecule–field state after accounting for the absorbed photon; changing the laser frequency shifts its effective energy relative to the open-channel threshold.

It is useful to separate the dimensional one-pole result from the additional assumptions made by this project. Eliminating one decaying molecular state in the renormalised zero-range model of Liu, Shi, and Wang gives [11]

$$a_c(E, I) = a_{bg} \left[1 + \frac{\Gamma_s(I)}{E - \nu_{re}(I) + i\gamma/2} \right]. \quad (14)$$

Here Γ_s is a *signed* resonant-strength parameter, while ν_{re} is the coupling-renormalised molecular energy. Equation (14) is already an isolated-resonance, zero-range description, but it does not by itself require the light shift $\nu_{re} - \nu$ to equal Γ_s .

The outline makes two further choices: threshold scattering, $E = 0$, and the parameterisation $\nu_{re}(I) = \nu + \Gamma_s(I)$. Microscopically, $\Gamma_s = m|\alpha_{re}|^2/(4\pi a_{bg})$ and therefore has the sign of a_{bg} . The implementation instead stores the ordinary non-negative magnitude $\Gamma = |\Gamma_s|$ and carries $s = \text{sgn}(a_{bg})$ explicitly, so that $\Gamma_s = s\Gamma$. Its dimensional map is consequently

$$a_c(I, \nu) = a_{bg} \left[1 + \frac{s\Gamma(I)}{-s\Gamma(I) - \nu + i\gamma/2} \right]. \quad (15)$$

For a positive background this reduces directly to the outline’s formula. For ^{88}Sr , $s = -1$: the code can retain the convenient bound $\Gamma \geq 0$ without changing loss into gain. Here $\Gamma(I) \propto |W|^2 \propto I$ is the weak-field stimulated-width magnitude, ν is the model detuning, and γ

is the molecular decay rate. All four quantities in the denominator must be expressed consistently as energies, angular frequencies, or ordinary frequencies; factors of 2π cannot be mixed. The real and imaginary parts of a_c remain linked by the same two controls and cannot be optimised independently [1]. Appendix A gives the two-channel derivation, algebraic checks, sign translation, and assumption ledger.

The identification $\nu_{re} - \nu = \Gamma_s$ is a simplifying control parameterisation, not a universal statement that an OFR light shift equals its resonance strength. In ^{88}Sr , the measured light-shift slopes even change sign between nearby 0_u levels, and an isolated-resonance expression fails once the detuning is comparable to the molecular-level spacing [8]. A calibrated model should therefore allow separate strength and shift functions before the optimised waveform is interpreted quantitatively.

Narrow intercombination transitions reduce the natural linewidth and can therefore produce appreciable scattering changes at moderate intensity. The early ytterbium demonstration changed the zero-energy scattering phase using this strategy [5]. More directly relevant here, optical Feshbach resonances have been measured in a thermal ^{88}Sr gas and used to control the collapse and expansion of an ^{88}Sr condensate [8, 9]. The small negative background scattering length makes even changes of a few Bohr radii dynamically important.

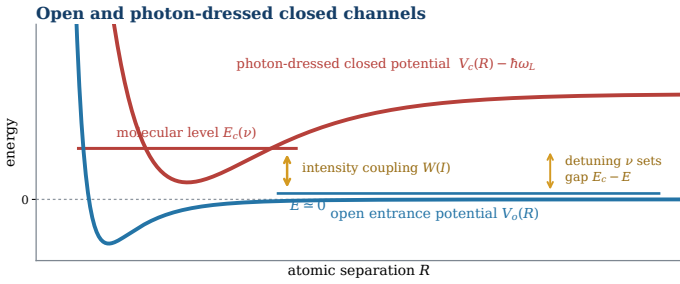


Figure 1. Two-channel energy picture for an optical Feshbach resonance, adapted conceptually from Ref. [7]. Here $V_o(R)$ is the open entrance potential and $V_c(R) - \hbar\omega_L$ is the photon-dressed closed potential. They support a near-zero-energy scattering state and molecular bound state E_c , respectively. Detuning controls the mismatch $E_c - E$, and optical intensity sets the coupling $W(I)$. The curves are schematic, not fitted $^{88}\text{Sr}_2$ potentials.

3. MODEL AND CONTROL PROBLEM

3.1. Dimensionless variables

Choose a fixed positive reference time t_* , independent of the as-yet unknown background scattering length. Its associated length is

$$t_* = \frac{m\ell_*^2}{\hbar}, \quad \ell_* = \sqrt{\frac{\hbar t_*}{m}}, \quad \tau = \frac{t}{t_*}, \quad \tilde{a} = \frac{a}{\ell_*}. \quad (16)$$

Optical frequencies are separately measured in units of γ :

$$u(\tau) = \frac{\Gamma(t)}{\gamma}, \quad v(\tau) = \frac{\nu(t)}{\gamma}. \quad (17)$$

The background ratio and its derived sign are

$$r_{bg} = \frac{a_{bg}}{\ell_*} = \sqrt{\frac{m}{\hbar t_*}} a_{bg}, \quad s = \text{sgn}(r_{bg}). \quad (18)$$

Equation (15) becomes the mapping implemented in the code,

$$\tilde{a}_c(t) = r_{bg} \left[1 + \frac{su(t)}{-su(t) - v(t) + i/2} \right]. \quad (19)$$

At $u = 0$, Eq. (19) reduces exactly to $\tilde{a}_c = r_{bg}$. The value $r_{bg} = 0$ is excluded because s would be undefined. For the present ^{88}Sr input, $r_{bg} = -0.008716$ and hence the resonant denominator is $u - v + i/2$; the low- and intermediate-cap controls in Fig. 2(c) visibly track $v \simeq u$. The admissible controls obey

$$0 \leq u(t) \leq u_{\max}, \quad |v(t)| \leq v_{\max}, \quad (20)$$

with $u_{\max} = \Gamma_{\max}/\gamma$.

Once a fixed t_* is selected, each candidate mass and background length maps to a definite r_{bg} through Eq. (18). This retains the physical dependence on m and a_{bg} rather than absorbing it into a species-dependent ruler.

With $t = t_*\tau$, $\eta = \ell_*\tilde{\eta}$, and $t_* = m\ell_*^2/\hbar$, the dimensional factor in Eq. (8) cancels. The dimensionless history-kernel coefficient required by the short-time derivation is therefore

$$\kappa_{th} = \frac{1}{8\pi^{3/2}\sqrt{i}}. \quad (21)$$

Unresolved solver convention

The continuous noninteracting derivation gives $\eta(0) = 0$ and the coefficient in Eq. (21). The current implementation instead initialises $\eta_0 = -4\pi\tilde{a}_c(0)$ and sets $\kappa_{code} = -1/(4\pi^{3/2}\sqrt{i})$. These coefficients differ in both sign and a factor of two. This is a known model audit, not a small plotting uncertainty; all optimisation results remain provisional until an independent benchmark selects the correct convention.

3.2. Molecular-yield objective

Under the convention in Eq. (19), the loss-contact relation gives a non-negative product flux proportional to $\text{Im}(1/a_c)|\eta|^2$. In the implemented dimensionless normalisation, the accumulated objective is

$$\mathcal{J}_m[u, v] = \frac{1}{2\pi} \int_0^T \text{Im}\left(\frac{1}{a_c(t)}\right) |\eta(t)|^2 dt. \quad (22)$$

This is the dimensionless, density-like quantity maximised in the current inelastic simulations. It is non-negative for either sign of r_{bg} .

Counting convention to resolve

Restoring the dimensional scales gives the density conversion $n_{\text{mol}} = g_2(0)\ell_*^3\mathcal{J}_m$, because $g_2(0)$ has units L^{-6} . The remaining physical derivation must establish whether this normalisation counts depleted atoms, lost pairs, or product molecules, and which pair factor applies to the chosen gas. A positive global factor does not change the optimising protocol, but it changes every quoted density, fraction, and molecule number. Figure 4 evaluates this conversion with the supplied $g_2(0) = 1 \text{ m}^{-6}$ and labels the result a *density proxy*; it is not an experimentally calibrated absolute yield until the counting convention is fixed.

3.3. Operational regularisation

Piecewise controls can exploit the discretisation by developing unrealistically sharp features. The current code penalises first and second differences of the *physical dimensionless controls* u and v (not cap-normalised controls). With $\Delta t = T/N$, it subtracts

$$\begin{aligned} \mathcal{R}_{\text{sm}} = & \frac{\lambda_{u,1}}{\Delta t} \sum_{k=0}^{N-1} (u_{k+1} - u_k)^2 + \frac{\lambda_{v,1}}{\Delta t} \sum_{k=0}^{N-1} (v_{k+1} - v_k)^2 \\ & + \frac{\lambda_{u,2}}{\Delta t^3} \sum_{k=0}^{N-2} (u_{k+2} - 2u_{k+1} + u_k)^2 \\ & + \frac{\lambda_{v,2}}{\Delta t^3} \sum_{k=0}^{N-2} (v_{k+2} - 2v_{k+1} + v_k)^2. \end{aligned} \quad (23)$$

The first-difference coefficients are configured through smoothness, and the second-difference coefficients through sharpness, with optional control-specific overrides. Because the amplitudes are not normalised by their caps, the same coefficient does not impose the same relative smoothing at different $u_{\max}$; this is one reason the coefficients were swept separately for each selected cap. The optimiser maximises the regularised score

$$\mathcal{S} = \mathcal{J}_m - \mathcal{R}_{\text{sm}}, \quad (24)$$

while all quoted molecular yields must report the unpenalised $\mathcal{J}_m$ as well. A posteriori slew tests are applied separately to avoid interpreting the regulariser as a hardware model.

4. NUMERICAL METHODS

4.1. Time discretisation and L1 history solver

The interval $[0, T]$ is divided into N equal steps, $t_k = k\Delta t$, with controls represented at $N + 1$ grid points. The L1 discretisation uses the history weights

$$w_{k,j} = \sqrt{k-j} - \sqrt{k-j-1}, \quad 0 \leq j \leq k-2. \quad (25)$$

The implementation initialises $\eta_0 = -4\pi a_c(t_0)$ and defines $q = 2\kappa_{\text{code}}/\sqrt{\Delta t}$ with $\kappa_{\text{code}} = -1/(4\pi^{3/2}\sqrt{i})$. For $k \geq 1$, let

$$H_k = \sum_{j=0}^{k-2} w_{k,j}(\eta_{j+1} - \eta_j), \quad (26)$$

the implemented recurrence has the explicit form

$$\eta_k = \frac{-1 + q(\eta_{k-1} - H_k)}{1/[4\pi a_c(t_k)] + q}. \quad (27)$$

Molecular yield is evaluated with the composite trapezoidal rule. A direct scan of the k earlier samples gives $O(N^2)$ arithmetic and $O(N)$ state storage for one control. Automatic differentiation has the same nonlocal dependency structure. The mismatch $\kappa_{\text{code}} \neq \kappa_{\text{th}}$ and the nonzero numerical initialisation are stated explicitly because optimiser stability cannot validate either choice.

4.2. Bounded physical controls

Unconstrained optimiser variables $x_k, y_k \in \mathbb{R}$ are mapped smoothly into the admissible laser domain:

$$u_k = u_{\text{max}} \text{sigmoid}(x_k), \quad v_k = v_{\text{max}} \tanh(y_k). \quad (28)$$

This parameterisation makes every iterate physical with respect to the amplitude bounds and remains differentiable for automatic differentiation. Saturation of either map can, however, reduce gradients; cap-usage diagnostics are therefore retained.

4.3. Gradient optimisation

The calculation follows the logic of GRAPE: discretise the controls, differentiate the terminal performance with respect to every time slice, and update the complete pulse [3]. JAX differentiates through the scattering map, Volterra scan, quadrature, and regulariser. Batched Adam updates are compiled and vectorised, with 64-bit arithmetic used for production calculations.

The learning rate is changed in stages without resetting Adam's moment estimates. Stability is evaluated at fixed checkpoint intervals independently of those stages. For run r , the reported solution is the best finite iterate according to $\mathcal{S}$ when smoothing is active and according to $\mathcal{J}_m$ otherwise.

4.4. Multi-start strategy and convergence tests

Initialisation is campaign-specific and is archived per run. The completed nine-cap sweep uses 50 reproducible five-mode Fourier starts at every cap. Each of the first smoothness-sharpness grids contains 6×5 coefficient pairs and 15 starts per pair: ten new Fourier controls plus five controls selected from earlier cap-matched results, for 450 completed runs per cap. The later 15×15 exploratory grids use ten Fourier starts and one queried control per coefficient pair. Targeted peak and L-BFGS stages then resume only the strongest saved controls. This staged design allocates broad sampling to discovery and long trajectories to refinement, but it also means that best values from different stages are methodology-dependent upper envelopes rather than identically distributed estimates.

The current stopping test is evaluated every M optimiser steps in the normalised feasible controls, not in the unbounded sigmoid/tanh coordinates. With $\bar{u} = u/u_{\text{max}}$ and $\bar{v} = v/v_{\text{max}}$,

$$\epsilon_u = \frac{\|\bar{u}_k - \bar{u}_{k-M}\|_2}{\sqrt{N+1}}, \quad \epsilon_v = \frac{\|\bar{v}_k - \bar{v}_{k-M}\|_2}{\sqrt{N+1}}. \quad (29)$$

Score stability uses the relative change between successive block-mean scores. A run is accepted only after the score and every enabled control metric pass their tolerances for three consecutive checkpoints; peak refinement can additionally require a projected-gradient threshold in

normalised control space. A saved best iterate is retained even when a wall-time limit or failure prevents this stopping rule from being met, which is why status is shown explicitly in Fig. 2.

Table 1. Completed fixed-method cap-sweep protocol. Every displayed value is recoverable from configuration document 1, queue 696482.

Quantity	Current value
Grid intervals N	100
Horizon and background	$T = 4, r_{\text{bg}} = -0.008716$
Intensity caps	10, 20, 40, 80, 160, 320, 640, 1280, 2560
Detuning bound	$v_{\text{max}} = 1000$
Adam parameters	$\alpha_0 = 0.05, \beta_1 = 0.9, \beta_2 = 0.999$
Stage schedule	(1000, 1), (10,000, 0.1), (100,000, 0.1)
Stability interval	500 steps
Tolerances	$J : 10^{-5}$; normalised $u, v : 10^{-3}$
Regularisation	smoothness 5×10^{-7} ; sharpness 5×10^{-8}
Initialisations	50 five-mode Fourier starts per cap
Arithmetic/device	JAX float64; CPU

Learning-rate multipliers compound, so the three rates are 0.05, 0.005, and 0.0005.

The schedule ends after 111,000 steps; this fixed campaign did not auto-halt individual runs.

4.5. Calibration sequence and reproducibility

The current methodology proceeds in a source-labelled sequence:

1. run a fixed-method nine-cap, 50-start baseline at $N = 100$;
2. scan smoothness and sharpness at the tight and high caps, and run a dedicated 250-seed crossover screen at $u_{\text{max}} = 320$;
3. polish every cap-320 seed and refine its top 20 controls while broadening the other promising coefficient regions;
4. refine the best saved controls with monotone peak search and L-BFGS; and
5. transfer identical controls to finer grids before interpreting the high-cap envelope.

Numerical data, immutable configuration documents, optimiser histories, and controls are stored separately from plotting code. The showcase generator writes both the figure and a JSON provenance record, so numerical claims can be checked without rerunning an optimiser.

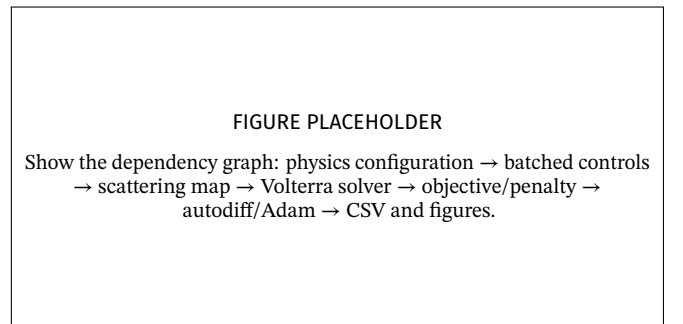


Figure 2. Computational workflow and artefact flow. Annotate the source-labelled directories used for configurations, results, figures, and logs.

5. VERIFICATION AND VALIDATION

5.1. Analytical checks of the scattering map

Equation (19) should be checked in three limits before optimisation: $u \rightarrow 0$ gives $a_c \rightarrow a_{\text{bg}}$; large $|v|$ suppresses the optical shift and loss; and the near-resonant response reproduces the expected dispersive real part and absorptive imaginary part. These checks also establish the sign conventions for v and $\text{Im } a_c$.

Table 2. Validation ledger at the current data snapshot. “Unit” denotes an automated implementation test; it is not a substitute for an independent physical benchmark.

Test	Expected behaviour	Result
No laser	$a_c/a_{bg} \rightarrow 1$	Pass (unit)
Far detuned	optical term $\rightarrow 0$	Pass (algebra)
Complex sign	$\text{Im } a_c \leq 0$	Pass (unit)
Constant a_s	reference $\eta(t)$	Open
Square-root ramp	universal onset law	Open
Grid refinement	convergent $\mathcal{J}_m$ and pulse	Fails at high cap
Autodiff gradient	finite-difference agreement	Open

5.2. Volterra solver and quadrature convergence

Optimisation stability at fixed N is not evidence of time-grid convergence. For a sequence such as $N = 100, 200, 400, 800, 1000$, define the relative yield difference

$$\epsilon_N = \frac{|\mathcal{J}_m^{(N)} - \mathcal{J}_m^{(N_{\text{fine}})}|}{\max(|\mathcal{J}_m^{(N_{\text{fine}})}|, 10^{-12})}. \quad (30)$$

If no analytic reference is available, the difference between the two finest grids is an error estimate, not a rigorous bound. Report the error in $\eta(t)$ where a reference is available and whether qualitative control features persist. Because optimisation can exploit grid-specific structure, use both routes: evaluate one fixed interpolated pulse on every grid, and independently reoptimise to comparable convergence at every N .

FIGURE PLACEHOLDER

Panel (a): real and imaginary parts of $\eta(t)$ for several N . Panel (b): relative yield error versus Δt on log axes with a fitted slope. Panel (c): optimised controls after interpolation to a common grid.

Figure 3. Validation and grid-convergence tests for the L1 Volterra solver and optimised objective. State the reference solution and error norm in the caption.

5.3. Gradient and baseline checks

On a reduced grid, compare automatic derivatives of $\mathcal{S}$ with centred finite differences for randomly selected control components. Then reproduce the real-scattering-length or unconstrained baseline associated with the universal onset problem before introducing the physical OFR mapping.

[TO COMPLETE] Give numerical tolerances, hardware, software versions, and the commit identifier. A statement such as “the curves look similar” is not a validation criterion; quote norms and relative errors.

6. RESULTS

6.1. Completed fixed-method cap sweep

The completed fixed-method campaign (configuration document 1, queue 696482) contains nine intensity caps, 50 Fourier starts per cap, and 111,000 Adam steps per start. Figure 4(a) shows every seed at every cap; no run is removed on the basis of its objective. At $u_{\text{max}} = 40$, the median and best values are 39.97 and 81.676; at 320 they are 65.84 and 1,043; and at 1280 they are 75.64 and 3,154, respectively. The

separation between median and best therefore grows by orders of magnitude and is dominated by rare optimisation trajectories.

The $u_{\text{max}} = 2560$ population does not continue this upper envelope. All 50 starts remain near $\mathcal{J}_m = 0.000572$ under the same schedule. This is an optimiser collapse associated with the fixed parameterisation and schedule, not evidence that molecule production physically vanishes above a threshold. Conversely, the large isolated best values at $u_{\text{max}} = 320$ –1280 cannot be interpreted as a saturation curve. The completed sweep establishes strong cap and seed sensitivity, but it does not by itself locate a crossover or plateau.

6.2. Regularisation search and targeted refinement

Separate archived cap-40 and cap-1280 searches include completed 6×5 smoothness-sharpness grids. Cap 320 instead uses a dedicated crossover screen: 250 broad Fourier seeds, a monotone polish of every seed, and loose L-BFGS refinement of the top 20. The stages contain 520 completed run records and expose strong seed dependence: their best objectives rise from 167.16 in exploration to 436.73 after all-seed polishing and 947.1 in the top-20 stage.

The cap-40 value $\mathcal{J}_m = 81.68$ agrees to the displayed precision across long Adam, peak-refinement, and L-BFGS records. The selected cap-320 endpoint has $\mathcal{J}_m = 947.1$, but its L-BFGS run ended at the time limit after 1,250 iterations rather than satisfying the stability test. Moreover, the fixed-method cap-320 campaign contains a still larger rare-seed objective, 1,043. The selected point is therefore a source-labelled control for comparison, not a converged cap-320 optimum or an absolute molecule yield.

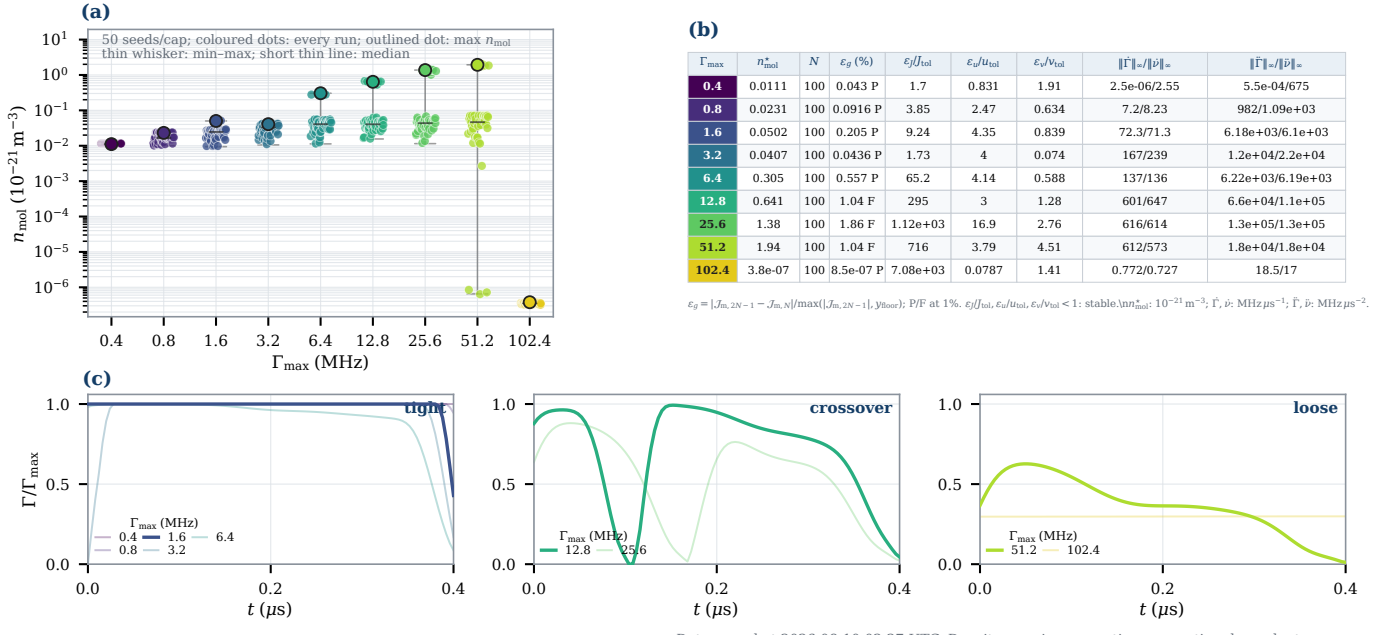
6.3. Control structure and the high-cap failure mode

Figure 4(c) now compares like with like: at each cap it takes the largest regularised-score run from the same completed fixed-method campaign and plots the archived $u(t)$ and $v(t)$ without dividing by u_{max} or v_{max} . The tight group contains caps 10–160, the crossover group contains 320 and 640, and the loose group contains 1280 and 2560. Separate vertical scales are used only between these three regimes, where the allowed amplitudes differ by orders of magnitude; within a regime the curves share one raw dimensionless scale and retain the cap colours used in panels (a) and (b). This presentation exposes the transition from predominantly cap-active tight controls to more structured crossover and loose-cap histories without conflating amplitude changes with normalisation.

The separately targeted cap-1280 trajectory is not substituted into that fixed-method comparison. Its latest saved iterate reaches the density proxy $n_{\text{mol}} = 5.34 \times 10^{-21} \text{ m}^{-3}$ after 10,605,750 steps, yet the objective is still increasing and none of the five strict cap-1280 members has passed the stopping rule. Its pulse contains narrow changes over one or a few grid intervals, including excursions near the endpoint. The regularisation penalty is finite, but it has not prevented the optimiser from using the $N = 100$ discretisation as an additional degree of freedom. The increasing objective must therefore be treated as a lower bound on a *discrete* optimisation trajectory, not as a physical optimum. Direct transfer of this fixed pulse from $N = 100$ to 800 changes the reevaluated objective by -5.43% , and the change from $N = 400$ to 800 is only $+0.08\%$. This shows that the forward evaluation of the interpolated pulse is settling; it does not show that a fresh high-cap optimisation would recover the same narrow structure or objective.

6.4. What is and is not established

The data establish three points. First, finite intensity matters: the tight-cap protocol is cap-active and has a repeatedly recovered saved objective near 81.68. Second, the optimisation landscape becomes strongly multimodal with increasing cap; population medians and best values answer different questions. Third, long high-cap searches



Data snapshot 2026-08-10 02:27 UTC. Density remains a counting-convention-dependent proxy.

Figure 4. Current numerical evidence, regenerated from archived data on 10 August 2026 UTC. (a) Dimensionless molecular objective J_m for all 450 fixed-method seeds across the full 9-cap campaign. Cap identity is colour coded across every panel. Thin whiskers span the minimum–maximum range, short thin bars give the median, and outlined points mark the run with the largest regularised score at each cap. (b) Fixed-control grid refinement: each of those nine best-score $N = 100$ pulses is linearly transferred and reevaluated at $N = 100, 200, 400, 800$ without reoptimisation. This tests the forward evaluation of the same pulse; it does not establish convergence of independently reoptimised optima. (c) Raw, unnormalised dimensionless controls $u(t)$ and $v(t)$ for the same selected runs, grouped into tight ($u_{\text{max}} \leq 160$), crossover ($u_{\text{max}} = 320, 640$), and loose ($u_{\text{max}} = 1280, 2560$) regimes; the horizontal axis is t/t_* . Physical interpretations elsewhere use $t_* = 10^{-7} \text{ s}$, $a_{\text{bg}} = -1.4a_0$, $m = 1.459707 \times 10^{-25} \text{ kg}$, $\gamma = 40,000 \text{ Hz}$, and $g_2(0) = 1 \text{ m}^{-6}$, but these conversions are deliberately not applied to this figure. Exact database paths, configuration document IDs, queue IDs, run IDs, statuses, and summary statistics are written to `figures/showcase_results_metadata.json`.

expose grid-scale structure rather than a plateau. Accordingly, no crossover cap, saturation yield, low-cap scaling exponent, or experimentally calibrated molecule fraction is reported here.

Figure 4(b) now separates one part of control optimisation from solver resolution. Between $N = 100$ and 800 , the fixed-control objective changes by $+0.32\%$, $+2.40\%$, and -5.43% at caps 40, 320, and 1280, respectively. Independent reoptimisations must still test whether each pulse structure and objective persist under matched physical regularisation. Only after that test, the kernel/initialisation audit, and the counting-prefactor derivation should initialise-seed and laser-calibration uncertainty be propagated into a physical yield.

7. EXPERIMENTAL TRANSLATION

7.1. Fixed representative: bosonic strontium-88

The physical reference is an ^{88}Sr condensate in the $^1S_0 + ^1S_0$ entrance channel, optically coupled near the $^1S_0 + ^3P_1$ molecular asymptote. The atomic intercombination line is near 689 nm; its measured ^{88}Sr frequency is 434 829 121 311(10) kHz [2]. The condensate experiment used the second-least-bound excited molecular level, with binding energy $h \times 24 \text{ MHz}$ below this asymptote [9]. The reduced-map baseline is $a_{\text{bg}} = -1.4(6)a_0 \simeq -7.41 \times 10^{-11} \text{ m}$; because this value is unusually close to zero, a sensitivity sweep over roughly $-2.0a_0$ to $-0.8a_0$ is more defensible than treating the central value as exact [8].

Two linewidths answer different questions. The ideal radiative molecule has $\gamma_{\text{mol}} = 2\gamma_a$ with $\gamma_a/2\pi = 7.5 \text{ kHz}$, hence $\gamma_{\text{mol}}/2\pi = 15 \text{ kHz}$. The direct thermal-gas OFR measurement instead fitted an effective total width $\gamma_{\text{eff}}/2\pi = 40(5) \text{ kHz}$, incorporating additional broadening [8]. The former is an ideal-theory input; the latter is the appropriate baseline for the present empirically broadened reference. The figures use ordinary frequencies with $\gamma_f = 40 \text{ kHz}$ throughout. An angular-frequency calculation is equivalent only if Γ , ν , and γ are all multiplied by 2π .

The measured intensity calibration is naturally expressed through

the optical length ℓ_{opt} rather than a universal constant ξ :

$$\ell_{\text{opt}} = \beta I, \quad \Gamma_{\text{stim}} = 2k_r \ell_{\text{opt}} \Gamma_{\text{mol}}, \quad \beta = 2.2 \times 10^4 \frac{a_0}{\text{W cm}^{-2}}, \quad (31)$$

where k_r is the relative collision wave number and $\Gamma_{\text{mol}} = 2\pi \times 15 \text{ kHz}$ [9]. If the code’s non-negative magnitude is identified as $\Gamma = \Gamma_{\text{stim}}$, while $\gamma = \Gamma_{\text{mol}}$ and $\nu = \Delta$, then

$$I_{\text{max}} = \frac{v_{\text{max}}}{2k_r \beta}, \quad \frac{\Delta(t)}{2\pi} = (15 \text{ kHz}) \nu(t). \quad (32)$$

Thus an intensity cap cannot be assigned until k_r is fixed consistently from the simulated gas. By contrast, under this ideal-width choice the current $v_{\text{max}} = 1000$ maps directly to $|\Delta|/2\pi = 15 \text{ MHz}$; the same dimensionless bound would be 40 MHz under the effective-width choice used in the showcase dimensionalisation. Both exceed the 10 MHz range quoted for the reference condensate experiment and must not be mixed in one conversion. More fundamentally, Eq. (31) calibrates a positive stimulated linewidth, whereas the tied shift in Eq. (15) remains a model assumption; the two identifications require validation against the selected molecular level.

7.2. Experimentalist’s recipe

Use $I = 0.057 \text{ W cm}^{-2}$, waist $725 \mu\text{m}$, and detunings of order 1 MHz as a literature-anchored comparison point, not yet as the optimised recommendation. The quoted beam corresponds to approximately 0.47 mW if I is the peak intensity of a Gaussian beam. Once the fixed reference time t_* has been selected, the reported 1.2- and 4-ms exposures must be converted using $\tau = t/t_*$; they cannot be represented by silently setting the solver’s $T = 1$ equal to one millisecond.

If the four reduced parameters a_{bg} , $\Gamma(I)$, ν , and γ have been calibrated in one convention, no separate C_6 or van der Waals length is needed to evaluate the algebraic map. Those microscopic quantities have already been compressed into its fitted inputs. They become necessary again when the isolated, zero-range reduction fails and a finite-range coupled-channel calculation is required. The minimum experimental data ledger is therefore the signed a_{bg} , a level-specific

Table 3. Fixed ^{88}Sr reference parameters. “Derived” values use the central reported quantities and are not additional measurements.

Parameter	Value	Source
Isotope and channel	Bosonic ^{88}Sr ; $^1S_0 + ^1S_0 \rightarrow ^1S_0 + ^3P_1$	[8, 9]
Atomic line	689 nm; 434 829 121 311(10) kHz	[2]
Background length	$a_{\text{bg}} = -1.4(6)a_0 \simeq -7.41 \times 10^{-11} \text{ m}$; sweep $-2.0a_0$ to $-0.8a_0$; the condensate paper uses the rounded value $-2a_0$	[8, 9]
Molecular reference	Second-least-bound level at -24 MHz ; ideal $\gamma_{\text{mol}}/2\pi = 15 \text{ kHz}$; measured effective $\gamma_{\text{eff}}/2\pi = 40(5) \text{ kHz}$	[8, 9]
Measured light shifts	0_u levels $n = -2, -3, -4$: respectively $-251(67)$, $+29(2)$, and $+4(1) \text{ kHz}/(\text{W cm}^{-2})$	[8]
Optical coupling	$\ell_{\text{opt}}/I = 2.2 \times 10^4 a_0/(\text{W cm}^{-2})$; at 0.057 W cm^{-2} , $\ell_{\text{opt}} = 1.25 \times 10^3 a_0$	[9]; derived
Demonstrated controls	$ \Delta /2\pi$ up to 10 MHz , or $ \Delta /\Gamma_{\text{mol}} \simeq 667$; typical $\alpha = 20a_0$ at $+1 \text{ MHz}$; changes up to $a_{\text{opt}}/a_{\text{bg}} = \pm 10$	[9]
Condensate	$N_0 \simeq 7000$, peak density $n_0 = 1 \times 10^{15} \text{ cm}^{-3}$, initial size $\sigma_0 = 0.8 \mu\text{m}$, trap mean frequency $2\pi \times (60 \pm 5) \text{ Hz}$	[9]
Laser and timing	Waist $725 \mu\text{m}$; example intensity 0.057 W cm^{-2} ; 1.2- and 4-ms exposures. The intensity corresponds to 0.47 mW for an on-axis Gaussian convention	[9]; derived
Loss scale	$K_{\text{in}} \sim 10^{-12} \text{ cm}^3 \text{ s}^{-1}$, millisecond sample lifetimes, fitted enhancement $\eta_{\text{loss}} = 19.5$	[9]

FIGURE PLACEHOLDER

Primary axis: predicted ^{88}Sr molecular yield/fraction versus peak 689-nm intensity, with a power axis for $w = 725 \mu\text{m}$. Mark 0.057 W cm^{-2} as the literature comparison and the optimised operating point separately. Inset: intensity and detuning waveforms in ms and MHz relative to the -24-MHz PA level.

Figure 5. Experimental prescription for ^{88}Sr . Propagate uncertainty in a_{bg} , ℓ_{opt}/I , density, and collision momentum; state whether the prediction is a dimensionless objective, density, fraction, or absolute molecule number.

strength calibration, an independently measured light shift, the appropriate total molecular width, and an unambiguous detuning origin.

The experimental isolated-resonance model contains the fitted loss enhancement $\eta_{\text{loss}} = 19.5$ and a detuning-dependent background loss not present in Eq. (15) [9]. The subscript distinguishes this empirical loss parameter from the pair amplitude $\eta(t)$ in Eq. (13). The ^{88}Sr prescription is therefore conditional on first verifying the mapping between the measured model and the implemented $a_c(u, v)$, especially the roles of Γ_{stim} , Γ_{mol} , and the phenomenological loss terms.

[TO COMPLETE] Calculate k_r for the final cloud model, convert every optimised $u(t)$ through Eq. (32), and compare the resulting peak power, modulation bandwidth, and duration with Table 3. Re-run the dynamics with the measured enhanced and background loss before claiming a quantitative ^{88}Sr molecule yield.

8. DISCUSSION

8.1. Physical interpretation

The low- and intermediate-cap controls suggest resonance tracking rather than a two-stage “prepare then convert” protocol. For $r_{\text{bg}} < 0$, increasing u shifts the resonant denominator unless v changes with it; the observed $v \simeq u$ relation compensates this shift while the intensity remains cap-active. The endpoint turn-off lowers both coupling and detuning together. Establishing a more detailed mechanism requires plotting $|\eta|^2$, instantaneous inelastic flux, and cumulative $\mathcal{J}_m$ from a grid-converged pulse. In particular, the present data do not justify invoking contact preparation or a quantum-Zeno mechanism.

The half-order memory dynamics are inherited from the universal short-time description of Ref. [10], but the reachable interaction

histories are not universal. The complex optical map, spontaneous linewidth, finite detuning range, cap-dependent regularisation, and bounded parameterisation introduce additional scales. Consequently, neither the real-scattering square-root ramp nor its scale-invariant growth law can be assumed to remain optimal in this lossy two-control problem; it must be evaluated as a common baseline after the solver convention is validated.

8.2. Limitations

The principal limitations affect different conclusions and should not be combined into one generic uncertainty bar.

- **Reduced optical map.** The calculation retains one s -wave entrance channel and one isolated, zero-range molecular state at $E = 0$, with constant Markovian decay, weak-field $\Gamma \propto I$, a uniform laser, and no medium correction. It additionally ties the light shift to the signed resonance strength, $\nu_{\text{re}} - \nu = \Gamma_s$. Finite temperature, saturation, power broadening, spatial intensity variation, neighbouring $0_u/1_u$ levels, and independently calibrated AC Stark shifts are therefore absent; Appendix A records the benefit and failure mode of each assumption.
- **Kernel and initial condition.** The exact noninteracting theory has $\eta(0) = 0$, whereas the recurrence starts from $-4\pi a_c(0)$ and uses a kernel coefficient that is not yet reconciled with Eq. (21). This can change both scale and pulse ranking. An analytic constant- a_s or square-root benchmark is required before production claims.
- **Finite time grid.** The high-cap optimiser develops one-interval features while $\mathcal{J}_m$ continues to rise, which is consistent with upward discretisation bias. The fixed-control transfer in Fig. 4(b)

is mildly grid-dependent and settles on its two finest grids, but matched reoptimisation across increasing N remains the direct test of the optimised envelope.

- **Regularisation model.** Equation (23) penalises absolute dimensionless controls, so its physical effect changes with the cap. A measured laser transfer function or bandwidth-constrained parameterisation would provide a defensible hardware model.
- **Local optimisation.** The widening best–median gap shows that rare basin discovery dominates the reported upper envelope. Larger seed populations improve discovery but cannot by themselves establish convergence to a global optimum.
- **Yield normalisation.** The missing counting prefactor does not change an optimiser under a positive global rescaling, but it prevents conversion of $\mathcal{J}_m$ into a molecule fraction or number.
- **Many-body and optical physics.** Depletion, trap inhomogeneity, finite range, the measured $\eta_{\text{loss}} = 19.5$ enhancement, and far-wing background loss are omitted. These omissions generally make a laboratory yield less favourable than the idealised two-body objective.
- **Intensity calibration.** The conversion $u \mapsto I$ depends on collision momentum k_r and therefore on the chosen cloud model. Until k_r and t_* are fixed, the dimensionless pulse has no unique power or duration.

8.3. Mathematical and computational extensions

The immediate computational extension is a grid ladder with identical source controls, not a more elaborate optimiser. Once a resolution-safe control basis is established, projected-gradient or L-BFGS polishing can replace millions of small Adam updates. A reduced Hessian spectrum can then quantify poorly constrained pulse directions, and robust control can optimise an ensemble over measured amplitude, detuning, and timing noise. These extensions become scientifically useful only after the forward Volterra solver has an independent benchmark.

9. CONCLUSION

A bounded optical Feshbach resonance produces a strongly cap-dependent control landscape because intensity changes both channel coupling and the resonance condition. At $N = 100$, the cap-40 pulse is repeatedly recovered with a saved dimensionless objective near $\mathcal{J}_m = 81.68$, and targeted search reaches 947.1 at cap 320, although that selected endpoint is time-limited. The cap-40 control tracks $v \simeq u$ within the assumed tied-shift map, whereas the cap-320 endpoint develops a distinct two-control history. At higher cap, population medians and rare best runs separate sharply, and the cap-1280 objective continues to grow through grid-scale pulse structure. Direct forward reevaluation of the fixed selected controls changes modestly by $N = 800$, but does not establish matched-grid optimisation convergence. The central answer is therefore conditional: finite optical power clearly shapes the reachable protocol, but the present data do not establish a crossover cap, a saturation yield, or an experimental molecule number. The next decisive steps are to reconcile the kernel and initialisation, reoptimise controls on the finer grids already used for fixed-pulse transfer, and only then restore the ^{88}Sr density, collision-momentum, and counting prefactors.

AUTHOR CONTRIBUTIONS

Charlie Mullen: software, numerical investigation, data curation, visualisation, and writing—original draft. Yangqian Yan: conceptualisation, methodology, supervision, and writing—review and editing.

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DATA AND CODE AVAILABILITY

The reusable package is under `src/ofc/`; immutable run documents are stored in `run_config/`, numerical arrays and metadata in `results/`, run logs in `logs/`, and notebook interfaces in `scripts/`. The report figures are regenerated by `reports/scripts/generate_figure1.py` and `reports/scripts/generate_showcase_results.py`. The second script also writes `showcase_results_metadata.json`, which records the exact database, configuration document, queue, run status, and summary values behind Fig. 4. A public repository URL or archival DOI and the final commit identifier should be added at submission.

A. ORIGIN, CHECKS, AND SCOPE OF THE OPTICAL MAP

This appendix makes explicit which parts of the optical map follow from a two-channel calculation and which parts are additional control-model assumptions. Throughout the derivation only, set $\hbar = 1$, so energies, detunings, linewidths, and rates have the same units. Factors of $\hbar$ are restored when dimensional experimental quantities are discussed.

A.1. Two-channel origin

Let $\hat{b}_k$ annihilate a ground-state atom and $\hat{d}_q$ an excited molecule. The open-channel contact strength is g , the laser-induced atom–molecule coupling is α , the bare molecular detuning is ν , the molecular decay rate is γ , the atomic mass is m , and the quantisation volume is Ω . A minimal Hermitian Hamiltonian is

$$\begin{aligned} \hat{H}_{\text{OFR}} = & \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} \hat{b}_{\mathbf{k}}^\dagger \hat{b}_{\mathbf{k}} + \sum_{\mathbf{q}} \left(\frac{g^2}{4m} + \nu \right) \hat{d}_{\mathbf{q}}^\dagger \hat{d}_{\mathbf{q}} \\ & + \frac{g}{2\Omega} \sum_{\mathbf{k}, \mathbf{k}', \mathbf{p}} \hat{b}_{\mathbf{k}+\mathbf{p}}^\dagger \hat{b}_{\mathbf{k}'-\mathbf{p}}^\dagger \hat{b}_{\mathbf{k}'} \hat{b}_{\mathbf{k}} \\ & + \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{k}, \mathbf{q}} \left[\alpha \hat{d}_{\mathbf{q}}^\dagger \hat{b}_{\mathbf{q}/2-\mathbf{k}} \hat{b}_{\mathbf{q}/2+\mathbf{k}} + \text{H.c.} \right], \end{aligned} \quad (33)$$

where $\epsilon_{\mathbf{k}} = k^2/(2m)$. The two collision paths are direct background scattering and optical conversion of an atom pair to the molecular state and back. In the two-body sector, irreversible molecular decay is represented by

$$\hat{H}_{2b} = \hat{H}_{\text{OFR}} - \frac{i\gamma}{2} \sum_{\mathbf{q}} \hat{d}_{\mathbf{q}}^\dagger \hat{d}_{\mathbf{q}}. \quad (34)$$

This is the no-jump two-body Hamiltonian associated with the molecular Lindblad decay; the imaginary energy $-i\gamma/2$ is not an additional coherent light shift [11].

Eliminating the molecule at relative collision energy E produces the energy-dependent open-channel interaction

$$U_{\text{eff}}(E) = g + \frac{|\alpha|^2}{E - \nu + i\gamma/2}. \quad (35)$$

The corresponding vacuum T matrix is

$$T(E) = [U_{\text{eff}}^{-1}(E) - \Pi(E)]^{-1}, \quad (36)$$

$$\Pi(E) = \frac{1}{\Omega} \sum_{\mathbf{k}} \frac{1}{E - k^2/m + i0^+}. \quad (37)$$

Because the contact model has no spatial range, $\Pi(E)$ is ultraviolet divergent. Thus g , α , and ν are bare quantities and cannot be inserted directly into a numerical prediction.

Match Eq. (36) to the low-energy s -wave convention

$$T(E) = \frac{4\pi}{m} \frac{1}{a^{-1}(E) + ik}, \quad k = \sqrt{mE}. \quad (38)$$

With a momentum cutoff Λ , the zero-range renormalisation can be

written

$$a_c(E) = a_{bg} + \frac{m|\alpha_{re}|^2}{4\pi} \frac{1}{E - \nu_{re} + i\gamma/2}, \quad (39)$$

$$\frac{m}{4\pi a_{bg}} = \frac{1}{g} + \frac{m\Lambda}{2\pi^2}, \quad (40)$$

$$|\alpha_{re}|^2 = \frac{|\alpha|^2}{[1 + m\Lambda/(2\pi^2)]^2}, \quad (41)$$

$$\nu_{re} = \nu - \frac{m\Lambda|\alpha|^2}{2\pi^2 + m\Lambda}. \quad (42)$$

The cutoff dependence is thereby absorbed into the three observable low-energy parameters. Defining the signed resonant strength

$$\Gamma_s(I) = \frac{m|\alpha_{re}(I)|^2}{4\pi a_{bg}} \quad (43)$$

turns Eq. (39) into the dimensional one-pole map

$$a_c(E) = a_{bg} \left[1 + \frac{\Gamma_s(I)}{E - \nu_{re}(I) + i\gamma/2} \right]. \quad (44)$$

Since $|\alpha_{re}|^2 \propto I$ in the weak-field regime, $|\Gamma_s| \propto I$. Equation (44) follows from the reduced two-channel model. The further identification

$$\nu_{re}(I) \equiv \nu + \Gamma_s(I) \quad (45)$$

is a particular parameterisation of the renormalised molecular energy. It is not forced by Eqs. (40)–(42), nor is it a general physical identity between an optical linewidth and an AC Stark shift.

Applying Eq. (45) and then the threshold limit $E \rightarrow 0$ yields the outline's formula,

$$a_c(I, \nu) = a_{bg} \left[1 + \frac{\Gamma_s(I)}{-\nu - \Gamma_s(I) + i\gamma/2} \right]. \quad (46)$$

Thus the $-\Gamma_s$ in the denominator records the tied-shift convention. It should not be read as evidence that every physical OFR has a light shift equal to its resonant strength.

A.2. Loss sign and the implementation convention

The microscopic definition in Eq. (43) makes Γ_s have the same sign as a_{bg} . This point matters for ^{88}Sr , where $a_{bg} < 0$ and therefore $\Gamma_s < 0$. The code avoids a signed intensity bound by defining

$$s = \text{sgn}(a_{bg}), \quad \Gamma = |\Gamma_s| \geq 0, \quad \text{quad}\Gamma_s = s\Gamma. \quad (47)$$

Substitution in Eq. (46) gives exactly the implemented dimensional map, Eq. (15). Hence the code constraint $0 \leq \Gamma \leq \Gamma_{\max}$ is a bound on a magnitude; it is equivalent to $|\Gamma_s| \leq \Gamma_{\max}$ in the microscopic convention.

To expose the elastic and lossy parts, write $G = \Gamma_s$, $\delta = \nu + G$, and $h = \gamma/2$. Then

$$\text{Re } a_c = a_{bg} \left[1 - \frac{G\delta}{\delta^2 + h^2} \right], \quad (48)$$

$$\text{Im } a_c = -a_{bg} \frac{Gh}{\delta^2 + h^2}. \quad (49)$$

With $a_c = a - ib$ and $b \geq 0$, one has

$$b = a_{bg} \frac{G\gamma/2}{(\nu + G)^2 + (\gamma/2)^2} \geq 0, \quad (50)$$

because $a_{bg}G = m|\alpha_{re}|^2/(4\pi) > 0$. Equivalently, the code's $G = s\Gamma$ guarantees $\text{Im } a_c \leq 0$ for either sign of a_{bg} . Constraining a *signed* $G \geq 0$ while also using $a_{bg} < 0$ would instead violate this condition; that is not what the present implementation does.

The dimensionless form follows with $u = \Gamma/\gamma$ and $v = \nu/\gamma$:

$$\frac{a_c}{a_{bg}} = 1 + \frac{su}{-v - su + i/2}, \quad (51)$$

$$\text{Re } \frac{a_c}{a_{bg}} = 1 - \frac{su(v + su)}{(v + su)^2 + 1/4}, \quad (52)$$

$$\text{Im } \frac{a_c}{a_{bg}} = -\frac{su/2}{(v + su)^2 + 1/4}. \quad (53)$$

For ^{88}Sr , $s = -1$ and the resonant denominator is $u - v + i/2$, explaining why a model-internal resonance-tracking trajectory satisfies $v \simeq u$.

A.3. Limiting-case checks

These identities are useful independent checks on both the algebra and the implementation.

- **No laser.** At fixed detuning, $\Gamma \rightarrow 0$ gives $a_c \rightarrow a_{bg}$.

- **Far from resonance.** For $|\nu + G| \gg \gamma$,

$$a_c \simeq a_{bg} \left[1 - \frac{G}{\nu + G} - i \frac{G\gamma}{2(\nu + G)^2} \right]. \quad (54)$$

If also $|\nu| \gg |G|$, the elastic correction falls as $1/\nu$ while loss falls as $1/\nu^2$.

- **No molecular decay.** For $\gamma = 0$,

$$a_c = a_{bg} \frac{\nu}{\nu + G}. \quad (55)$$

The zero is at $\nu = 0$ and the pole at $\nu = -G$. In particular, the fraction is $\nu/(\nu + G)$, not its reciprocal.

- **Finite molecular decay.** When $\gamma > 0$, no real detuning produces a pole. The absorptive response is largest at $\delta = \nu + G = 0$, while the pole lies off the real detuning axis.

A.4. Assumption ledger

Table 4 separates the computational benefits of the reduction from the corresponding limits on interpretation. The map is attractive for optimal control precisely because it is algebraic, differentiable, and ties elastic response to loss. Those virtues do not make its high-intensity or far-detuned extrapolation quantitative.

A.5. ^{88}Sr substitution and detuning origin

Once the reduced parameters are known, evaluating the map requires no separate van der Waals table. A minimal input set is the signed background length, molecular decay rate, level-specific mapping from intensity to Γ , light-shift calibration, detuning origin, and one consistent frequency convention. For the central $a_{bg} = -1.4a_0$ value and the code's positive magnitude $\Gamma_f \geq 0$, Eq. (15) becomes

$$\frac{a_c}{a_0} = -1.4 \left[1 - \frac{\Gamma_f}{\Gamma_f - \nu_f + i\gamma_f/2} \right], \quad (56)$$

where all subscript- f quantities are ordinary frequencies. Use $\gamma_f/2 = 7.5 \text{ kHz}$ for an ideal 15 kHz molecular line or $\gamma_f/2 = 20 \text{ kHz}$ for the central experimentally fitted 40 kHz line. The same equation may instead be evaluated in angular frequency after multiplying *every* denominator term by 2π .

In the signed microscopic notation the shifted detuning is $\delta = \nu + \Gamma_s$; in the code notation it is $\delta = \nu + s\Gamma$. An experimental detuning may instead be quoted from the zero-intensity molecular line, the intensity-shifted line, the atomic $^1S_0 \rightarrow ^3P_1$ threshold, or a fitted photoassociation centre. These are not interchangeable. The measured 0_u shift slopes for $n = -2, -3, -4$ are respectively about $-251(67)$, $+29(2)$, and $+4(1) \text{ kHz/(W cm}^{-2})$ [8]; their variation directly demonstrates why the tied shift in Eq. (45) must be tested rather than assumed during quantitative experimental translation.

The practical conclusion is twofold. The present map is a useful simplified theoretical control model: it has the correct qualitative background, dispersive, resonant-loss, and far-detuned structures and is well suited to automatic differentiation. A mathematically optimal trajectory becomes a proposed ^{88}Sr laser protocol only after the strength, light shift, linewidth, detuning origin, temperature range, and high-intensity response have all been calibrated for one selected molecular level.

B. NONDIMENSIONALISATION AND CONVENTIONS

[TO COMPLETE] Complete the remaining contact and product-counting normalisation audit. The optical-map sign and Hz-versus-rad/s conventions are derived in Appendix A; the unresolved issue here is the prefactor that converts lost-pair flux into counted product molecules.

Table 4. Physical assumptions in the implemented optical map.

Assumption	Physical content	Benefit	Limitation
Two channels / one isolated state	One atomic entrance channel couples to one excited molecular level.	One resonance denominator; transparent gradients and interpretation.	Nearby 0_u and 1_u levels and continua interfere. For ^{88}Sr the approximation holds only when detuning is small relative to the molecular-level spacing; otherwise coupled channels are required.
s -wave scattering	Only the $\ell = 0$ partial wave is retained.	Correct threshold channel for sufficiently cold identical bosons.	Higher partial waves can contribute as temperature or collision energy rises.
Zero collision energy	$E \rightarrow 0$ and one complex length characterises scattering.	Removes an energy dimension from the optimisation.	Omits thermal averaging and effective-range corrections; a finite-temperature calculation must evaluate Eq. (44) over the collision-energy distribution.
Zero range	Molecular potentials are replaced by contact couplings.	Fast, analytic, differentiable, and inexpensive in JAX.	C_6 , R_{vdW} , E_{vdW} , and the free-bound Franck-Condon wavefunction are not explicit. They are compressed into fitted parameters and re-enter if those parameters are not known or the reduction fails.
Constant Markovian decay	The molecular energy contains a fixed $-i\gamma/2$.	Produces a complex scattering length and irreversible loss automatically.	Omits energy-, intensity-, time-, and detuning-dependent decay, technical laser broadening, predissociation, collisions, and non-Markovian effects.
Weak-field scaling	$ \Gamma_s = \kappa I$ with constant $\kappa > 0$.	Turns a laser-power limit into $0 \leq \Gamma \leq \Gamma_{\text{max}}$.	Saturation, power broadening, nonlinear AC Stark shifts, neighbouring-level coupling, and intensity-dependent decay are absent.
Shift tied to strength	$\nu_{\text{re}} - \nu = \Gamma_s$ uses one parameter for coupling strength and the resonance-centre shift.	Leaves only the two controls $\Gamma(t)$ and $\nu(t)$.	A general OFR has separately calibrated resonant strength and light shift; measured ^{88}Sr shifts are strongly level dependent and even change sign.
Uniform laser	Every pair experiences one intensity, detuning, and polarisation.	Produces a single control waveform.	A trapped cloud samples spatially varying intensity and density; beam propagation and polarisation impurity are ignored.
Vacuum two-body physics	No depletion or density-dependent medium correction modifies the scattering map.	Clean low-density connection between laser controls and a_c .	Can fail in a dense, depleted, or strongly interacting gas; the yield functional then needs a self-consistent many-body treatment.

Table 5. Symbol and unit ledger for the final derivation.

Symbol	Meaning	Unit/scale
a_{bg}	background scattering length	$-1.4(6)a_0$
a_c	complex scattering length	length; scale $\ell_\star$
γ	molecular decay rate	showcase: 40,000 Hz; ideal: $2\pi \times 15$ kHz in angular units
Γ	non-negative stimulated-width magnitude	$2k_r \ell_{\text{opt}} \Gamma_{\text{mol}}$; same frequency convention as γ
Γ_s	signed microscopic resonant strength	$s\Gamma$ with $s = \text{sgn}(a_{\text{bg}})$
ν	model detuning from the selected PA level	same frequency convention as γ
$\eta(t)$	short-range pair amplitude	length; scale $\ell_\star$
$\mathcal{C}$	Tan-contact density	inverse length ⁴
$\mathcal{J}_m$	current yield objective	dimensionless; prefactor unresolved

C. DISCRETE SOLVER DERIVATION

[TO COMPLETE] Derive the $L1$ weights and recurrence in Eqs. (25)–(27). State how the endpoint singularity is handled, why the newest increment enters the denominator, and the memory/time complexity of the scan.

D. ADDITIONAL OPTIMISATION DIAGNOSTICS

[TO COMPLETE] Place full per-seed convergence plots, zero-yield diagnostics, raw- versus bounded-control stability, alternative schedules, raw-space line scans, smoothness sensitivity, and higher-resolution cap-sweep tables here. Keep only decision-relevant summaries in the main text.

E. REPRODUCTION CHECKLIST

1. Record the repository commit and Python environment.
2. Freeze the learning-rate, smoothness, convergence, and initialisation settings.
3. Save run-level CSV/JSON data before plotting.
4. Label every result by grid, seed, cap, and configuration.
5. Regenerate report figures from saved numerical artefacts.
6. Confirm that all reported headline values can be read from an archived table.

F. FIXED REFERENCE TIME AND SWEEPABLE BACKGROUND RATIO

A reference time $t_\star > 0$ is fixed across candidate physical setups. It defines a positive length $\ell_\star = \sqrt{\hbar t_\star / m}$, not a removable physical scattering length. Consequently

$$r_{\text{bg}} = \frac{a_{\text{bg}}}{\ell_\star} = \sqrt{\frac{m}{\hbar t_\star}} a_{\text{bg}} \quad (57)$$

remains a signed model parameter. Sweeping r_{bg} represents candidate values of a_{bg} and m at the fixed $t_\star$.

With $s = \text{sgn}(r_{\text{bg}})$, the code evaluates

$$\tilde{a}_s = r_{\text{bg}} \left[1 + \frac{su}{-su - v + i/2} \right] = |r_{\text{bg}}| \left[s + \frac{u}{-su - v + i/2} \right]. \quad (58)$$

The prefactor is $|r_{\text{bg}}|$ because r_{bg} itself is signed; this guarantees the far-detuned or zero-width limit $\tilde{a}_s = r_{\text{bg}}$. The value $r_{\text{bg}} = 0$ is invalid because s would be undefined, and there is no independent sign parameter in the configuration.

Finally, the present regulariser is applied to first and second differences of the absolute dimensionless controls u and v , with powers of the dimensionless time step shown in Eq. (23). It is therefore not invariant under a change of control cap or duration; this cap dependence is one reason the regularisation search is treated as part of the

numerical methodology rather than as a calibrated hardware transfer function.

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